

EcoRig: A Ground-Serviceable Rigging System for Arboreal Camera Traps

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Abstract— Arboreal camera traps are essential for canopy biodiversity monitoring but remain hazardous and costly to service because they traditionally require climbing for battery/SD maintenance and view realignment. EcoRig rearchitects the service model by enabling ground-based maintenance after installation. The system comprises: (i) an adaptive strap with a tripod head that accommodates tree growth and slanted trunks; (ii) a non-backdrivable pan-tilt module (worm + planetary gearing) under ESP32 control for remote view alignment; (iii) a magnetic docking mechanism actuated by a pneumatic release; and (iv) an Edge-driven autocalibration routine that self-discovers mechanical limits without encoders or limit switches. Field tests with the National Parks Board Singapore demonstrated a 93.75% reduction in climbs (from 8/year to 1 every two years), service cycles ≈ 12 min, pan-tilt repeatability $\pm 0.7^\circ$, non-backdrivable hold $0.42 \text{ kg}\cdot\text{m}$ ($\sim 4.1 \text{ N}\cdot\text{m}$), WiFi latency ≈ 210 ms, and robust operation in tropical weather. While the 120-day battery life target was not met (21 days achieved), the results validate ground servicing feasibility, safety, and usability. This contribution lies in integrating adaptive mechanics with ML calibration to deliver a safe, modular, and scalable conservation tool.

Index Terms—arboreal camera traps, canopy monitoring, rigging system, tropical biodiversity

INTRODUCTION

Arboreal camera traps are essential for canopy wildlife monitoring because many species live primarily in trees. However, they are expensive and hazardous to maintain. Firms such as Camphora Pte. Ltd. spend nearly \$800 to service just five traps, while arborists from the National Parks Board Singapore (NParks) must climb manually, a process that is tiring, time-consuming, and dangerous. Maintenance is also frequent—every 1–6 months—due to the need for battery replacement and SD card retrieval. Existing rigging systems are bulky, vulnerable to insect nests, and still require climbing for adjustments and strap loosening. These limitations raise costs, compromise safety, and lead to inactive cameras when servicing is delayed, causing data loss and reduced monitoring reliability.

The proposed system addresses these burdens by enabling ground-based servicing of camera traps after installation. Beneficiaries include arborists, who face fewer occupational risks; researchers and conservation organizations, who benefit from reduced costs and greater operational efficiency; and ecosystems, which experience less disturbance. The broader impact lies in improving arborist safety, lowering financial expenditure, and ensuring continuous biodiversity data collection through longer camera uptime.

Arboreal camera trapping has gained increasing importance as researchers expand biodiversity studies into canopy habitats. Tree climbing remains the dominant servicing method, but requires extensive training, certification costs

exceeding \$2,000, and exposes climbers to safety hazards. Some alternative systems have emerged: the CanopyCam [1], is a partial rigging system that allows retrieval from the ground, though it still requires climbing for angle adjustment and strap loosening. Autonomous climbing robots such as Treebot (CUHK) [2] and RiSE v3 (UPenn) [3] demonstrate technological potential, but their high complexity, limited payload, and poor adaptability render them impractical for large-scale use.

These studies highlight a gap: no existing solution provides full ground-serviceability while remaining lightweight, non-damaging to trees, and resistant to weather and wildlife interference. Addressing this gap requires balancing operational efficiency, arborist safety, and ecological protection. By avoiding autonomous robots and instead focusing on a ground-serviceable rigging system, the present work offers a practical, cost-efficient, and deployable approach. Early testing validates this direction, showing successful trials with NParks, weatherproofing resilience, and secure strap mechanics that accommodate tree growth

I. PROPOSED APPROACH

A. System Requirements

The EcoRig system was defined by both stakeholder needs (NParks, Camphora) and technical constraints identified through literature review and prototyping. Success is defined by a reduction in climb frequency, while maintaining safety, reliability, and ecological sustainability.

Requirements were established through:

1. Problem analysis – identifying inefficiencies, safety hazards, and ecological risks with current practices.
2. Stakeholder engagement – insights from Camphora and NParks researchers highlighted limits on assembly weight need for pitch/roll tolerance on slanted trunks, and misalignment tolerances for docking.
3. Benchmarking – evaluating existing rigging systems and robotics solutions for feasibility.
4. Iterative testing – validating parameters through lab and field trials (e.g., waterproofing, strap tension, docking strength, wireless range). Field trials were conducted at the Singapore Botanic Gardens over the period of July and August 2025.

TABLE I. SYSTEM REQUIREMENTS

Subsystem	Requirement	Target (per EcoRig)
Overall system	Camera trap servicing time	< 20 min
	Installation/retrieval	< 30 min
	Corrosion resistance	No rust after 730 days outdoor exposure
	Impact of nesting	Nesting does not diminish functionality
	Carry-in weight	< 15 kg
	Static sag	< 1° total drift after 30 days
Adaptive strap	Minimum deployable trunk/branch diameter	20 cm
	Tree growth accommodation	12 cm growth over 2 years without constriction
	Strap expansion	+17 mm under tangential quasi-static load of 260 N
Tripod head	Tripod adjustment range	Pitch: -35° to +75°, roll: ±90°
	Tripod torque resistance	Holds > 33 Nm without slipping
Mount and magnetic dock	Mount strength	< 5° deflection and no plastic deformation after suspension of 10 kg load for 10 min
	Docking force	>15 kg vertical holding force, < 1.5 kg via release cord
	Misalignment tolerance	Successful dock with ±1 mm lateral, ±5° roll and pitch error
	Tug/shock resistance	No undocking or damage with sudden load of 8 kg with 5 m/s downward velocity
Pulley system	Line retention	0 derailments after 30 cycles
	Wear resistance	No sheath damage after 100 cycles
Weatherproof housing and electronics	Ingress protection	IP65 resistance
	WiFi control	Range >40 m LOS, >30 m in canopy, command latency < 250 ms
Pan and tilt gear system	Angular range	±90° pan, ±30° tilt
	Repeatability	<1° error over 10 repeats to same setpoint
	Resistance to backdrive torque	Gear system holds > 0.35 kg without slip under defined tamper load

B. Concept Selection

Several alternative approaches to enabling ground-serviceable arboreal camera trap systems were initially generated and evaluated. These included: legged climbers, gripper-based mechanisms, aerial platforms, rope-based rigging systems, and human Single Rope Technique (SRT), a conventional arborist practice where operators ascend and descend trees using a single static rope with mechanical ascenders. Each concept was assessed against a set of weighted criteria derived from stakeholder requirements: ease of use,

novelty, ascent speed, safety, cost of manufacturing, adaptability, opportunities for AI integration, and overall feasibility.

A Pugh chart was employed to systematically compare and rank the concepts relative to a baseline.

TABLE II. CONCEPT SELECTION PUGH CHART

Criteria	Weight (100%)	Concept Variants				
		Legged	Gripper	Aerial	Rope	Human SRT
Ease of Use	15	5	5	3	4	2
Novelty	5	5	5	4	4	1
Ascent Speed	5	2	1	5	3	3
Safety	20	4	4	3	4	1
Cost of Manufacturing	10	1	1	3	4	5
Adaptability	15	1	3	3	3	4
AI Opportunity	5	3	3	3	3	1
Feasibility	25	1	1	2	3	5
Total		2.55	2.8	2.9	3.5	3.1
Rank		5	4	3	1	2
Continue?		No	No	Maybe	Yes	-

The results revealed that:

- Legged climbers offered high novelty and potential ease of use but performed poorly in terms of feasibility, adaptability, and manufacturing cost, leading to the lowest overall ranking.
- Gripper devices demonstrated marginally higher feasibility but remained constrained by low adaptability and ascent speed, placing them second lowest.
- Aerial platforms scored highly on novelty but were penalised for excessive cost, limited feasibility, and safety concerns; they were considered a tentative “Maybe” for continuation.
- Rope-based rigs presented the most balanced performance, achieving strong scores in safety, adaptability, and cost. This variant was ranked the highest overall and recommended for further development.
- Human SRT achieved strong marks for feasibility and cost, reflecting its established use in arboriculture. However, it ranked poorly in safety, ease of use, and adaptability, as it still necessitates repeated climbing and thus does not eliminate the primary operational hazard.

The Pugh chart analysis indicated that rope-based rigs provided the most promising foundation for meeting the project’s objectives. While other options demonstrated isolated

strengths, they were ultimately eliminated either due to impracticality (legged climbers and grippers), excessive complexity and cost (aerial platforms), or insufficient mitigation of safety concerns (human SRT). Consequently, the design process converged on a rope-based concept, subsequently refined through the integration of adaptive strapping, modular sub-assemblies, a motorised pan-tilt mechanism, and magnetic docking with pneumatic release, culminating in the final EcoRig system.

C. Concept Description

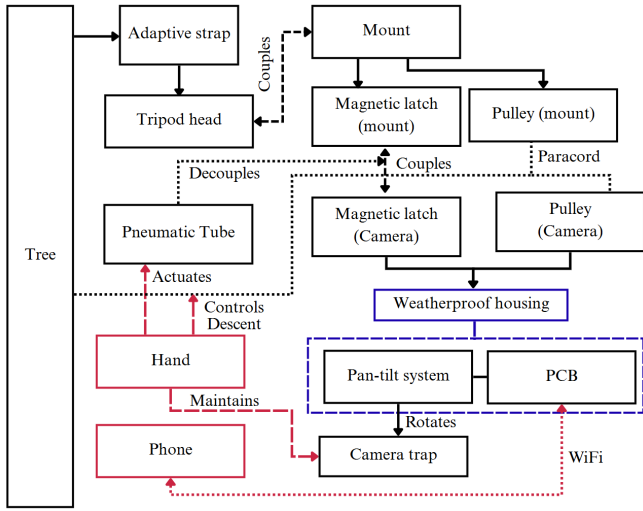


Figure 1. System architecture of the EcoRig concept.

System Architecture. The EcoRig system is composed of several integrated subsystems that together enable safe, ground-based servicing of arboreal camera traps. At the highest level, the EcoRig comprises a mount that is permanently secured to the tree via an adaptive release strap. A pulley system allows the camera module to be raised and lowered from the ground, while magnetic coupling and release mechanisms ensure reliable docking and retrieval. The camera module itself houses a waterproof enclosure containing the trap, control electronics, and a motorised pan-tilt system. Alignment is performed remotely over Wi-Fi using an ESP32 microcontroller and web application live view, while battery and SD card changes are carried out manually at ground level.

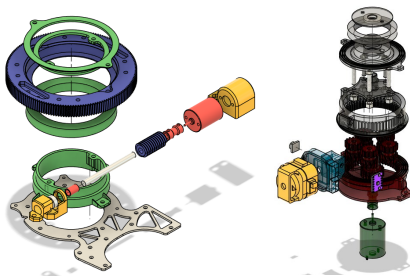


Figure 2. Exploded views of two actuation systems: (left) pan-tilt module with worm gear drive; (right) split-ring compound planetary gearset.

Actuation Systems. Two actuation architectures were considered. A worm-gear-driven pan-tilt module was selected for its non-backdrivable property, which prevents drift or

tampering by wildlife. In parallel, a split-ring compound planetary gearset was developed and evaluated for its compactness and higher gear reduction potential. While the planetary gearset offered efficiency, the worm gear proved more robust and reliable in field conditions, and thus was chosen for the final design.

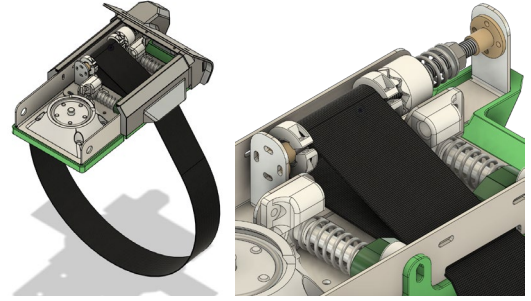


Figure 3. CAD renders of the adaptive release strap: (left) full assembly view; (right) detailed view of the internal ratcheting mechanism.

Adaptive Release Strap. The adaptive strap anchors the EcoRig to the tree trunk. It incorporates a ratcheting mechanism that provides dual behaviour: allowing slow, quasi-static expansion as the tree grows while resisting impulsive disturbances such as animal tugs. This prevents both constriction damage to the tree and accidental loosening of the mount, thereby ensuring long-term ecological compatibility and structural security.

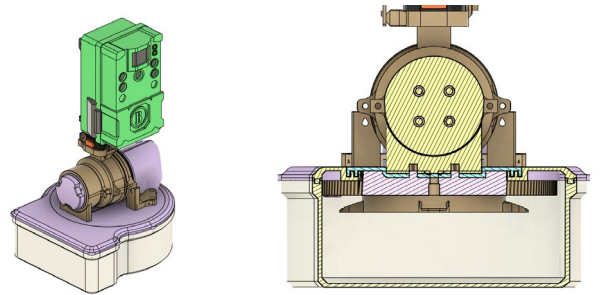


Figure 4. CAD renders of the waterproof housing box with labyrinth seal: (left) full assembly view; (right) cross-sectional view.

Waterproof Housing. The camera housing is designed to protect internal electronics under tropical field conditions. A labyrinth seal was primarily used and integrated into the housing profile, forcing fluids and contaminants to follow a tortuous path before reaching internal components. This eliminates the need for conventional gaskets, which degrade over time, and provides reliable ingress protection against both rain and insects.

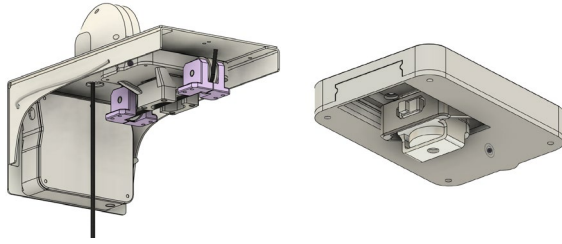


Figure 5. CAD renders of the pulley top mount: (left) full assembly view; (right) detailed view highlighting the magnetic release locking mechanism.

Pulley Top Mount. The top mount integrates sheaves sized to prevent derailment of the retrieval line during repeated service cycles. It also houses the magnetic release mechanism, which locks the camera module in place during operation but can be disengaged from the ground via pneumatic actuation. This combination of secure holding and controlled release ensures that the system remains both safe and easily serviceable.

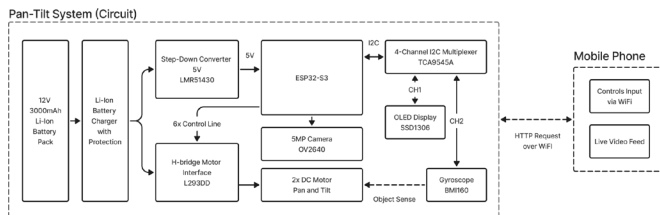


Figure 6. Circuit diagram & PCB layout for control systems

Electronics and Control. Power is supplied by a 12V 3000mAh battery, stepped down via a DC-DC converter to provide regulated 5V for the ESP32 microcontroller and sensors as seen in Fig. 6. The peak current draw was constrained to $\leq 3.5 \text{ A @ } 12 \text{ V}$ (requirement). Field measurement confirmed that motor surges remained within this envelope. However, battery endurance was 21 days, falling short of the 120-day target, mainly due to continuous H-bridge leakage.

The ESP32 integrates wireless connectivity, enabling operators to adjust the pan and tilt motors remotely through a web interface. The low-power deep-sleep capabilities make it suitable for field deployments as well.

At the core, the ESP32-S3 executes motor control, sensor fusion, calibration routines, and Wi-Fi hosting. It interfaces with a 5MP OV2640 camera module for live video streaming and an OLED display (SSD1306) for local diagnostics. Peripheral communications are managed through a TCA9545A I²C multiplexer, which allows multiple devices such as the OLED and BMI160 gyroscope to share the bus without address conflicts.

Instead of traditional encoders, EcoRig employs a BMI160 6-axis IMU (3-axis gyro + 3-axis accelerometer). The IMU enables proprioception of the pan-tilt mechanism: angular displacement, rate of change, and stall detection. This sensor supports both the edge computed auto-calibration routine and periodic drift and alignment checks. The use of gyroscope significantly reduces hardware usage.

A custom printed circuit board (PCB) consolidates motor drivers, power distribution, and signal routing, enabling compact and reliable integration within the waterproof housing.

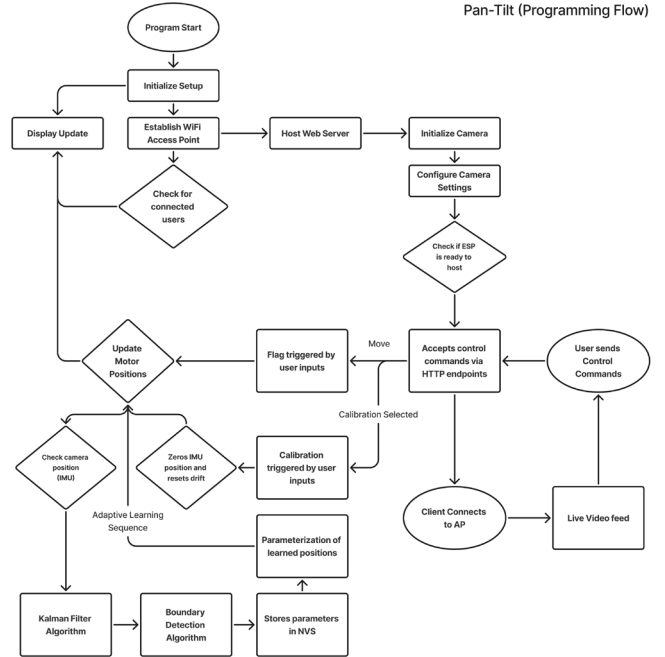


Figure 7. Programming flow of Pan-Tilt and Calibration Control

Programme Flow. EcoRig software was written and developed in C++ with three primary goals: reliability, adaptability, and convenience.

Controls:

- Low-level motor controls uses a Pulse-Width Modulation (PWM) to control the H-bridge drivers.
- Sensor fusion via the BMI160 gyro/accelerometer data is filtered using Kalman Filter algorithm.
- Finite State Machine (FSM) is used as a supervisory control for the calibration algorithm.

Web Application:

- Wi-Fi SoftAP mode creates a local wireless access point.
- Hosts a HTTP Server endpoint which is lightweight and runs directly from the ESP32S3.
- This also allows for any smartphone to connect to the network without any additional hardware.
- Live video feed is streamed over the same network for a seamless experience.

Intelligent Calibration Algorithm:

- Stall detection: Motors sweep until gyro angular velocity ≈ 0 under current draw spike, indicating a hard stop.
- Back-off & re-test: The system retracts slightly, then repeats to confirm limit.
- Map usable range: Limits are stored in flash memory, generating a digital map of pan/tilt range.

- Compensation: If asymmetry is detected (e.g., tilt $+38^\circ$ but -31°), the center point is auto-adjusted.

EcoRig's calibration relies on a lightweight Kalman filter to fuse gyroscope and accelerometer data from the BMI160, balancing noise rejection with computational efficiency on the ESP32-S3. Mechanical stall points are detected when the motor's angular velocity approaches zero while current spikes indicate resistance, defining hard stops without encoders or limit switches. The algorithm then performs a back-off and re-test cycle to confirm each limit, mapping the usable pan-tilt envelope and storing it in non-volatile memory. This approach ensures repeatable alignment within $\pm 0.7^\circ$ while automatically compensating for asymmetry between positive and negative limits.

Overall, the electronics and software integrates power management, motor control, sensing, and wireless communication into a compact, modular platform. This architecture enables EcoRig's key features: ground-based servicing, remote view alignment, and intelligent calibration without the need for additional encoders or limit switches.

II. MODELLING

A. Component Selection

Tilt Actuation System. The tilt actuation system was identified as a critical subsystem, since it must withstand both static and dynamic loads imparted by camera mass, wind forces, and potential tampering by wildlife. A worm gear (1:160) and a planetary gearset (1:171.5) were considered as the main reduction stages, driven by RS370 DC motors through an L293DD H-bridge and controlled by an ESP32S3 microcontroller with feedback from a BMI160 IMU.

The modelling objective for the planetary stage was to evaluate whether its gear teeth could sustain bending stresses corresponding to worst-case loading conditions, including the scenario of an 8 kg monkey exerting torque on the camera bracket. The Lewis equation was adopted to estimate tooth root bending stresses:

$$\sigma = \frac{F_t}{b \cdot m \cdot Y} \cdot K_o \cdot K_v \cdot K_m \quad (1)$$

Where σ is bending stress, F_t the tangential load, b the face width, m the module, and Y the Lewis form factor. The correction factors K_o , K_v , and K_m account for overload, dynamic effects, and load distribution respectively.

Key assumptions included:

- Only 2 of 5 planets share tangential load at any instant.
- The gears were manufactured in Bambu Lab ASA filament, with an allowable bending stress derated to 8 MPa to account for variability in FDM printing.
- Geometry: 15 teeth, module 1.03 mm, face width 7.5 mm, 20° full-depth spur tooth profile.
- Operating condition: $\pm 45^\circ$ swing with a 125 mm lever arm.

To reduce modelling uncertainty, the Lewis form factor Y was obtained using an AI-assisted tool that correlated gear geometry and material method to empirical form factors, recommending a conservative value of $Y=0.28$.

Adaptive Release System. The adaptive release system would ideally be a strap made of a material that can plastically deform in a controlled manner to maintain tension while accommodating to tree trunk growth. Such a material would involve advanced material science but instead, we could use a mechanical system to simulate it while the strap material remains inextensible. Springs to maintain tension while a toothed ratchet system to enable permanent lengthening of strap.

Two models were used to determine the spring constants and length required for springs, and another for the compression spring of the ratchet system.

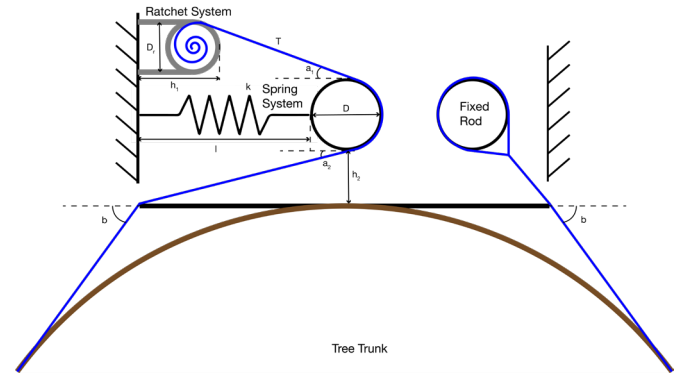


Figure 8. Simplified Diagram of Adaptive Release System

Firstly, spring system is modelled from Figure 7 to show a relationship between tension of strap T and spring constant, k_s and free length, l_f .

$$k_s(l_f - l) = \frac{1}{2}T \left[\left(\frac{l_1 + \frac{D}{2} - h_1}{\sqrt{D_f^2 + (l_1 + \frac{D}{2} - h_1)^2}} \right) + \left(\frac{l_1 + \frac{D}{2}}{\sqrt{h_2^2 + (l_1 + \frac{D}{2})^2}} \right) \right] \quad (2)$$

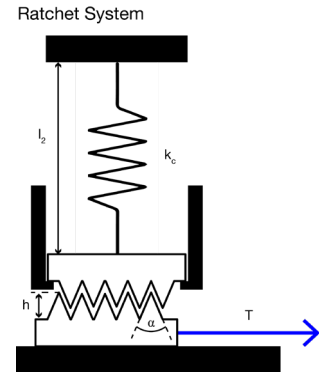


Figure 9. Simplified Diagram of Ratchet System

$$F_s = k_c(l_c - l_2) \quad (3)$$

$$F_s = T \left[\frac{\left(\mu \cos \left(\frac{\alpha}{2} \right) + \sin \left(\frac{\alpha}{2} \right) \right)}{\left(\mu \sin \left(\frac{\alpha}{2} \right) + \cos \left(\frac{\alpha}{2} \right) \right)} \right] \quad (4)$$

Secondly, the ratchet system is modelled based on Figure 8. Equation (3) and (4) shows the relationship between the tension, T that the ratchet can maintain on the strap and the needed spring constants, k_c and free length, l_c . To determine T needed, we will investigate that under Section B.

B. Performance Predictions

Tilt Actuation System. The model outputs bending stress predictions under different load scenarios. The baseline case predicted that at a horizontal hold ($\sin\theta=1$), tooth stress increased linearly with hanging mass. At a $\pm 45^\circ$ swing, the effective torque was reduced ($\sin\theta=0.707$), lowering peak stresses.

Results indicated:

- At 3.65 kg load, bending stress ≈ 3.65 MPa.
- At 5.17 kg load, bending stress ≈ 5.17 MPa.
- At 7.94 kg load, bending stress approached 7.94 MPa, close to the 8 MPa allowable limit.

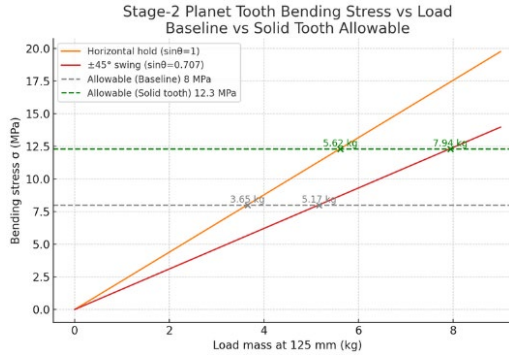


Figure 10. Predicted bending stress at the gear tooth root for the stage-2 planetary gearset under increasing load at 125 mm lever arm.

Thus, the planetary stage met requirements up to ~ 8 kg loading, satisfying the target of resisting tampering by medium-sized monkeys. A comparison against a solid-tooth allowable (≈ 12.3 MPa) further confirmed sufficient margin under normal service conditions.

Model predictions were validated experimentally by subjecting the assembled tilt system to shock and quasi-static loads, confirming that tooth failure did not occur within the expected service envelope. This demonstrated that the selected planetary stage was structurally adequate and that the worm gear could provide the additional self-locking functionality needed to ensure non-back-drivability.

Adaptive Release System. There are two conditions that have to be fulfilled. A, a force consideration and B, a length consideration.

Firstly, the force consideration is involved with the amount of tension needed in the strap. The upper limit, T_g is decided by the growth pressure of a typical tree trunk. Using concepts of hoop stress, we derived the following equation.

$$T_g = P_g r w \quad (5)$$

Where P_g is the growth pressure of a tree, r is the radius of the tree and w is the width of the strap used.

The lower limit of the tension is given by the tension required to hold up the weight of the system and the weight of

the largest animal in the trees. Around 8kg of additional loading.

$$T_m = \frac{(M_s + M)g}{2\mu \sin b} \quad (6)$$

Where M_s is the mass of system, M is the mass of a monkey, μ is the friction coefficient between the bark and the TPU cover and b can be referenced from Figure 7.

Hence the following condition (7) must be fulfilled to ensure system can survive monkey disturbance while not constricting tree growth. However, there are major assumptions on the conditions of tree growth, so to give the maximum possible margin, T must be kept as low as possible which equal to T_m with a safety factor.

$$T_m < T < T_g \quad (7)$$

TABLE III. VARIABLES TO DETERMINE LIMITS

Constants							
P_g	r	w	M_s	M	g	μ	b
0.5 [4]	0.1	0.05	4.5	8	10	0.6	55
Limits							
T_g				T_m			
2500				384			
Independent Variable: T = 400							

TABLE IV. VARIABLES FOR MODEL

Independent Variable: $T = 400$							
Constants							
D	D_r	l_1	l_2	h_1	h_2	α	μ
0.008	0.03	0.05	0.02	0.033	0.008	84	0.5
Dependent Variables							
k_s		l_f		k_c		l_c	
4.3		0.05		9.3		0.02	

III. RESULTS

A. Prototypes



Figure 11. Installation of strap and pulley mount on a slanted tree trunk.

Using the magnetic coupling of the tripod head, we achieved a modular assembly of the strap and pulley mount. This meant light (<1.6 kg) assembly groups and easy assembly without tools.



Figure 12. Levelling of housing using tripod head and spirit level to accommodate slanted tree trunk.



Figure 13. Successful docking using pulley system.



Figure 14. Fully assembled prototype on slanted tree trunk.

B. Experiments

Through experimentation, we endeavored not only to ensure that the prototype is safe and robust, but to ensure that it would surpass even unforeseen failure modes encountered during outdoor deployment. The experimentation included:

Structural Load & Impact Test. The assembled bracket–strap–tripod system was subjected to a sustained service load and sudden load impulses. Torques were applied to the bracket and strap hardware to represent worst-case, in-use knocks.

Adaptive (Slow-Rise) Loading. A ratchet strap was used to emulate the quasi-static increase of force that occurs in the field i.e. the gradual growth of the tree.



Figure 15. Adaptive Loading Testing in SUTD FabLab

Misalignment Tolerance. The mount was deliberately set at off-axis pitch and roll angles to assess functionality when installed on uneven or imperfect surfaces.

Weathering & Ingress. The system was left outdoors through adverse weather.

A live prototype test was also conducted in collaboration with NParks researchers. The prototype was installed at a height of 15 m on the trunk of a *Hopea odorata* on 5 August 2025, and was retrieved on 11 August 2025. All subsystems remained fully functional after two thunderstorms.



Figure 16. Prototype testing at Plant Resource Centre, NParks.

TABLE V. EXPERIMENTAL RESULTS

Subsystem	Requirement	Experimental result (green = target met)
Overall system	Camera trap servicing time	11:56 for 15m tree (2:15 untie, 0:21 descent, 2:32 battery + SD + settings, 1:33 ascent, 3:57 bundle cords, 1:18 pan-tilt)
	Installation/retrieval	12:31 for 15m tree (2:14 bag 1, 2:57 strap + tripod, 3:27 bag 2, 3:53 rigging + throwing) / 8:24 for 15m tree (0:25 descent, 4:21 untying, 3:38 bag down)
	Corrosion resistance	Visible rust after 8 days outdoor exposure
	Impact of nesting	0 nests observed during pilot
	Carry-in weight	≤ 4.5 kg packed kit (15m deployment)
	Static sag	0.8° drift after 8 days outdoor exposure test
Adaptive strap Tripod head	Strap expansion	Expands 1.5 mm under 520 N, ≤ 2 mm extension at 8 kg @ 5 m/s
	Tripod adjustment range	Adjustment range confirmed: -35° to $+75^\circ$ pitch, 360° roll, $+90^\circ$ yaw
	Tripod torque resistance	Holds 34 N·m, sag 0.7°

Mount and magnetic dock	Mount strength	Deflection 3° with 10 kg suspended for 10 min, no further deflection for suspensions for longer periods, no plastic deformation
	Docking force	9.3 kg vertical; release at 1.2 kg
Mount and magnetic dock	Misalignment tolerance	Dock success at $\pm 15^\circ$ pitch, $\pm 3^\circ$ roll error
	Tug/shock resistance	No undock/damage at 8 kg @ 5 m/s
Pulley system	Line retention	0 derailments over 50 cycles
	Wear resistance	No sheath damage after 100 cycles + 8 days sun/rain
Weatherproof housing and electronics	Ingress protection	IP35 pass (spray nozzle test; no ingress)
	Wi-Fi control	LOS: 46 m; canopy: 27 m; latency 210 ms
Pan and tilt gear system	Angular range	Pan $\pm 90^\circ$, Tilt $\pm 38^\circ$
	Repeatability	$\pm 0.7^\circ$ over 10 repeats
Pan and tilt gear system	Resistance to back drive torque	Holds 0.42 kg·m without slip

IV. DISCUSSION

Safety and Convenience. The modular assembly allowed for a fast and safe installation and retrieval, without the need for any tools. Servicing time was 75% shorter than the traditional service time of 40 minutes with manual climbing.

Weather Resistance. The weather-resistant housing was able to protect all electronics from two thunderstorms. More robust testing could be conducted to determine the exact IP rating of the system.

Tree Growth. The adaptive strap was able to expand under a quasi-static load, allowing the system to accommodate for tree growth during the deployment cycle.

Versatility. The tripod head, in conjunction with the pan and tilt gear system, allows the system to be deployed at any angle of tree trunk or branch at varying circumferences, while maintaining an ideal view angle at a predetermined observation site. One limitation is that the vertical path from the mount to the ground must not be densely obstructed by branches or leaves, though our prototype managed to overcome some obstruction by foliage.

Macaque Resistance. Our load-testing results and analytical gearset predictions show that our system can withstand the typical activity of wildlife and macaques, without sustaining critical damage or notable changes in view angle. This activity includes swinging/climbing on cords, landing on the housing, and deliberate tugging at the individual components.

Stakeholder Insights. We collaborated closely with NParks researchers - we conducted 2 formal interviews, observed 2 practice climb sessions, conducted a live prototype

test with installation and retrieval being 6 days apart, and a final prototype review at the project exhibition. At the live prototype test and final prototype evaluation, the researchers verbally commented that the system is convenient, safe, feasible, and potentially effective. Our results thus extend beyond empirical data and include the real user's positive feedback.

One suggested improvement from the NParks researchers would be to address "freak" macaque tampering incidents, such as them completely chewing through straps or cords. Existing solutions include using bicycle locks to secure the camera trap after the strap is torn, but this reintroduces the restriction of tree growth.

Long-term Limitations. Due to limitations in our project timeline, we were unable to experimentally determine the long-term performance of our prototype, and simulate a full deployment cycle of 2 years. Although we were able to make predictions through modelling, future experiments should still be conducted to assess real long-term performance.

V. CONCLUSIONS

EcoRig demonstrates that ground-serviceable, edge-driven adaptive arboreal monitoring is practical and beneficial. The system met or exceeded core mechanical goals such as self-lock, pose envelope, docking tolerance, shock survival and operational goals (service time, latency), and validated the adaptive strap concept for growth accommodation. The main shortfall is power longevity, earmarked for the next iteration via radio duty-cycling, deep sleep, and hardware power-gating. The approach generalizes to other tree-mounted or elevated field systems (e.g., sensor pods, acoustic arrays) where safe ground servicing is valuable.

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